

Functional & Non Functional Requirements

Team - 3 Hostel Management System

Team Members

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1. Functional Requirements

Priority:-

High = the app doesn't work without it.

Medium = important but the core flow doesn't depend on it.

Low = nice to have.

FR-1 Login and Roles

Priority: High

- FR-1.1 A user logs in with username and password, wrong credentials show error.
- FR-1.2 After login, a Student goes to the student dashboard and an Admin goes to the admin dashboard.
- FR-1.3 A Student trying to open an Admin page (even by typing the URL) gets blocked.
- FR-1.4 Logout ends the session.

FR-2 Student Registration

Priority: High

- FR-2.1 A new student registers with SRN, name, email, phone number and password.
- FR-2.2 Registration is rejected if the SRN or email is already used.
- FR-2.3 A student can edit their own contact info but not their SRN.
- FR-2.4 Admin can view and edit any student's record.

FR-3 Block and Room Setup

Priority: High

- FR-3.1 Admin can add a hostel block (name, number of floors).
- FR-3.2 Admin can add a room under a block with room number, floor, capacity, type (single/double/triple) and fee.
- FR-3.3 Two rooms in the same block can't share a room number.
- FR-3.4 A room with a student currently in it can't be deleted.

FR-4 Room Allocation

Priority: High

- FR-4.1 A student can apply for a room, optionally picking a block/type they'd prefer.
- FR-4.2 Admin sees a list of pending applications and can allot a vacant room to a student.
- FR-4.3 A room already at full capacity can't be allotted to another student.
- FR-4.4 A student who already has a room can't be given a second room.
- FR-4.5 Allotment updates the room's occupant count and the student's record together.
- FR-4.6 Admin can deallocate or move a student to a different room.
- FR-4.7 The student sees their new room on their dashboard as soon as it's allotted.

FR-5 Fee Tracking

Priority: Medium

- FR-5.1 The system stores fee due and fee paid per student per term.
- FR-5.2 Admin records a payment (amount + date)
- FR-5.3 A payment can't be recorded as more than what's still owed.
- FR-5.4 A student can see only their own fee status.

FR-6 Complaints

Priority: Medium

- FR-6.1 A student raises a complaint with a category and description.
- FR-6.2 A new complaint starts as "open" and is logged with the student's room and time.
- FR-6.3 Admin can move it to "In Progress" and then "Resolved". Only Admin can close a complaint, not the student who raised it.
- FR-6.4 A student sees only their own complaints while the Admin sees all of them.

FR-7 Visitor Log

Priority: Low

- FR-7.1 A visitor entry records name, phone number, purpose, host student, and entry time
- FR-7.2 Admin marks a visitor as checked out, which records the exit time.
- FR-7.3 A visitor can't be logged against a student who doesn't currently have a room.

FR-8 Admin Dashboard

Priority: Medium

- FR-8.1 Shows total rooms, occupied rooms, and occupancy percentage.
- FR-8.2 Shows how many students owe fees and the total amount owed.
- FR-8.3 Shows how many complaints are Open or In Progress.
- FR-8.4 Shows how many room applications are pending.

FR-9 Search and Filter

Priority: Medium

- FR-10.1 Admin can search students by name or SRN (partial match works).
- FR-10.2 Admin can filter rooms by block, floor, type and status.
- FR-10.3 Admin can filter complaints by status and category.
- FR-10.4 No matches shows a "no results found" message, not an error.

2. Non-Functional Requirements

NFR-1 Usability

The people using this day to day are hostel staff, not developers, so it has to be usable without training. Applying for a room, raising a complaint, and recording a payment should each take no more than two clicks from the dashboard. Form errors should show next to the field that's wrong, in plain language and shouldn't wipe out what the user already typed.

NFR-2 Performance

Pages (room list, dashboard, search) should load within 2 seconds under normal use.

NFR-3 Security

Passwords are hashed, never stored as plain text. Every Admin-only page checks the role on the server, not just by hiding a link in the UI. A student changing an ID in the URL to view someone else's fee record or complaint should be blocked, not shown the data.

NFR-4 Reliability

If two Admins try to allot the same last free bed in a room at the same time, only one should succeed and the other should get a clear rejection and the room should never end up over capacity. Any action that touches more than one table (like allotment) should either fully complete or not happen at all.

NFR-5 Scalability

The system should keep working with more students and rooms added over time without needing a redesign with roughly up to 1000 students and 500 rooms without rewriting anything.

NFR-6 Maintainability

Code is split into separate Django apps - accounts, rooms, fees, complaints, visitors , so one person's module doesn't get tangled up with another's. Each app has its own basic tests.

NFR-7 Portability

Should run on different operating systems and browsers.

NFR-8 Availability

The system should be up during hostel office hours (9 am–6 pm on working days) with minimal downtime. If something crashes, the user sees a plain error page, not a stack trace.

3. Project Distributions

Member	Features	Responsibilities
Shalik S (Lead)	FR-1, FR-2, FR-8	Login security - hashed passwords, role checks on every admin route
Shreyas Patil	FR-3, FR-4	No double-booking a room, even under concurrent use
Shreyas Mohanty	FR-5, FR-9	Reports match the current database and load fast
Shobhit Rachit Keshri	FR-6, FR-7	Complaint status updates show up immediately, no stale data